

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ cd buildroot/
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release/buildroot$ ls
arch      CHANGES      COPYING      envsetup.sh  Makefile.legacy  scripts      utils
archives  Config.in      DEVELOPERS   fs           package          support
board     Config.in.legacy  dl.tar.gz    linux        README          system
boot      configs       docs         Makefile     README.rockchip  toolchain
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release/buildroot$ tar -xzf dl.tar.gz
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release/buildroot$ ls
arch      CHANGES      COPYING      docs          Makefile      README.rockchip  toolchain
archives  Config.in      DEVELOPERS   envsetup.sh   Makefile.legacy  scripts          utils
board     Config.in.legacy  dl           fs           package        support
boot      configs       dl.tar.gz    linux        README          system
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release/buildroot$
```

图 12

1.4 系统开发环境配置

如需进行 U-Boot、内核开发，请在 LinuxSDK 源码目录下，执行如下命令，配置环境变量。

备注：配置系统开发环境变量前，请勿执行配置应用环境变量，否则会导致系统镜像编译失败。

```
Host# export PATH=/home/tronlong/RK3576/rk3576_linux6.1_release/prebuilts/gcc/linux-
x86/aarch64/gcc-arm-10.3-2021.07-x86_64-aarch64-none-linux-gnu/bin/:$PATH
```

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ export PATH=/home/tronlong/RK3576/rk3576_linux6.1_release/prebuilts/gcc/
linux-x86/aarch64/gcc-arm-10.3-2021.07-x86_64-aarch64-none-linux-gnu/bin/:$PATH
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$
```

图 13

```
Host# aarch64-none-linux-gnu-gcc -v
```

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ aarch64-none-linux-gnu-gcc -v
Using built-in specs.
COLLECT_GCC=aarch64-none-linux-gnu-gcc
COLLECT_LTO_WRAPPER=/home/tronlong/RK3576/rk3576_linux6.1_release/prebuilts/gcc/linux-x86/aarch64/gcc-arm-10.3-2021.07-x86_64-aarch64-none-
linux-gnu/bin/./libexec/gcc/aarch64-none-linux-gnu/10.3.1/lto-wrapper
Target: aarch64-none-linux-gnu
Configured with: /data/jenkins/workspace/GNU-toolchain/arm-10/src/gcc/configure --target=aarch64-none-linux-gnu --prefix= --with-sysroot=/a
aarch64-none-linux-gnu/libc --with-build-sysroot=/data/jenkins/workspace/GNU-toolchain/arm-10/build-aarch64-none-linux-gnu/install/aarch64-
none-linux-gnu/libc --with-bugurl=https://bugs.linaro.org/ --enable-gnu-indirect-function --enable-shared --disable-libssp --disable-libmud
flap --enable-checking=release --enable-languages=c,c++,fortran --with-gmp=/data/jenkins/workspace/GNU-toolchain/arm-10/build-aarch64-none-
linux-gnu/host-tools --with-mpfr=/data/jenkins/workspace/GNU-toolchain/arm-10/build-aarch64-none-linux-gnu/host-tools --with-mpc=/data/jenk
ins/workspace/GNU-toolchain/arm-10/build-aarch64-none-linux-gnu/host-tools --with-isl=/data/jenkins/workspace/GNU-toolchain/arm-10/build-a
aarch64-none-linux-gnu/host-tools --enable-fix-cortex-a53-843419 --with-pkgversion='GNU Toolchain for the A-profile Architecture 10.3-2021.07
(arm-10.29)'
Thread model: posix
Supported LTO compression algorithms: zlib
gcc version 10.3.1 20210621 (GNU Toolchain for the A-profile Architecture 10.3-2021.07 (arm-10.29))
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$
```

图 14

因我们的存在，让嵌入式应用更简单

备注：执行如下命令完成应用开发环境变量配置后，当前 Terminal 命令行终端将不再适用 U-Boot、内核开发。如需进行 U-Boot、内核开发，请另开一个 Terminal 命令行终端进行相关工作。

Host# source /home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/host/environment-setup

```
tronlong@tronlong-virtual-machine:~$ source /home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/host/environment-setup

[Buildroot]

Making embedded Linux easy!

Some tips:
* PATH now contains the SDK utilities
* Standard autotools variables (CC, LD, CFLAGS) are exported
* Kernel compilation variables (ARCH, CROSS_COMPILE, KERNELDIR) are exported
* To configure do "./configure $CONFIGURE_FLAGS" or use the "configure" alias
* To build CMake-based projects, use the "cmake" alias

tronlong@tronlong-virtual-machine:~$
```

图 19

执行如下命令，查看交叉编译工具链版本信息。

Host# aarch64-buildroot-linux-gnu-gcc -v

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ aarch64-buildroot-linux-gnu-gcc -v
Using built-in specs.
COLLECT_GCC=/home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/host/bin/aarch64-buildroot-linux-gnu-gcc.
br_real
COLLECT_LTO_WRAPPER=/home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/host/libexec/gcc/aarch64-buildroot-linux-gnu/12.4.0/lto-wrapper
Target: aarch64-buildroot-linux-gnu
Configured with: ./configure --prefix=/home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/host --sysconfdir=/home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/host/etc --enable-static --target=aarch64-buildroot-linux-gnu --with-sysroot=/home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/host/aarch64-buildroot-linux-gnu/sysroot --enable-cxx --enable-cxx-abi-compat --with-gnu-ld --disable-libssp --disable-multilib --disable-decimal-float --enable-plugins --enable-lto --with-gmp=/home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/host --with-mpc=/home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/host --with-mpfr=/home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/host --with-pkgversion='Buildroot 2024.02' --with-bugurl=https://gitlab.com/buildroot.org/buildroot/-/issues --without-zstd --disable-libquadmath --disable-libquadmath-support --enable-tls --enable-threads --without-isl --without-cloog --with-abi=lp64 --with-cpu=cortex-a72.cortex-a53 --enable-languages=c,c++ --with-build-time-tools=/home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/host/aarch64-buildroot-linux-gnu/bin --enable-shared --disable-libgomp
Thread model: posix
Supported LTO compression algorithms: zlib
gcc version 12.4.0 (Buildroot 2024.02)
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$
```

图 20

2. Linux 系统镜像编译、生成

因我们的存在，让嵌入式应用更简单

2.1 配置编译选项

2.1.1 配置 LinuxSDK 编译选项

进入 LinuxSDK 源码目录, 配置 LinuxSDK 编译选项。

Host# ./build.sh lunch:tl3576_evm_defconfig

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ ./build.sh lunch:tl3576_evm_defconfig
Log colors: message notice warning error fatal

Parsing supported commands...

Log saved at /home/tronlong/RK3576/rk3576_linux6.1_release/output/sessions/2024-12-30_17-45-23
Switching to defconfig: /home/tronlong/RK3576/rk3576_linux6.1_release/device/rockchip/.chip/tl3576_evm_defconfig
mkdir -p /home/tronlong/RK3576/rk3576_linux6.1_release/output/kconf/lxdialog
make CC="gcc" HOSTCC="gcc" \
  obj=/home/tronlong/RK3576/rk3576_linux6.1_release/output/kconf -C /home/tronlong/RK3576/rk3576_linux6.1_rele
ase/device/rockchip/common/kconfig -f Makefile.br conf
make[1]: Entering directory '/home/tronlong/RK3576/rk3576_linux6.1_release/device/rockchip/common/kconfig'
gcc -D DEFAULT_SOURCE -D XOPEN_SOURCE=600 -DCURSES_LOC="" -DNCURSES_WIDECHAR=1 -DLOCALE -I/home/tron
long/RK3576/rk3576_linux6.1_release/output/kconf -DCONFIG_="" -MM *.c > /home/tronlong/RK3576/rk3576_linux6.1
_release/output/kconf/.depend 2>/dev/null || :
gcc -D DEFAULT_SOURCE -D XOPEN_SOURCE=600 -DCURSES_LOC="" -DNCURSES_WIDECHAR=1 -DLOCALE -I/home/tron
long/RK3576/rk3576_linux6.1_release/output/kconf -DCONFIG_="" -c conf.c -o /home/tronlong/RK3576/rk3576_linu
x6.1_release/output/kconf/conf.o
gcc -D DEFAULT_SOURCE -D XOPEN_SOURCE=600 -DCURSES_LOC="" -DNCURSES_WIDECHAR=1 -DLOCALE -I/home/tron
long/RK3576/rk3576_linux6.1_release/output/kconf -DCONFIG_="" -I. -c /home/tronlong/RK3576/rk3576_linux6.1_re
lease/output/kconf/zconf.tab.c -o /home/tronlong/RK3576/rk3576_linux6.1_release/output/kconf/zconf.tab.o
gcc -D DEFAULT_SOURCE -D XOPEN_SOURCE=600 -DCURSES_LOC="" -DNCURSES_WIDECHAR=1 -DLOCALE -I/home/tron
long/RK3576/rk3576_linux6.1_release/output/kconf -DCONFIG_="" /home/tronlong/RK3576/rk3576_linux6.1_release/
output/kconf/conf.o /home/tronlong/RK3576/rk3576_linux6.1_release/output/kconf/zconf.tab.o -o /home/tronlong/RK
3576/rk3576_linux6.1_release/output/kconf/conf
rm /home/tronlong/RK3576/rk3576_linux6.1_release/output/kconf/zconf.tab.c
make[1]: Leaving directory '/home/tronlong/RK3576/rk3576_linux6.1_release/device/rockchip/common/kconfig'
#
# configuration written to /home/tronlong/RK3576/rk3576_linux6.1_release/output/.config
#
Using current kernel version(6.1)
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$
```

图 21

2.1.2 配置设备树

我司提供的设备树文件位于 LinuxSDK 源码"kernel/arch/arm64/boot/dts/rockchip/"目
录, 具体说明如下。

表 4

文件名称	说明
tl3576-evm.dts	基础设备树, 支持 TL3576-EVM 评估板基本外设, 不支持 LVDS L CD 显示。
tl3576-lvds-1024x768.dts	基于 tl3576-evm.dts 设备树增加支持 LVDS LCD 显示, 但不支持 MIPI LCD 显示。

评估板使用的设备树文件存在多级的依赖关系, 具体如下表所示。表中的下一级设备
树文件, 被上一级设备树文件所引用。

因我们的存在, 让嵌入式应用更简单

表 5

tl3576-lvds-1024x768.dts			
tl3576-evm.dts			
rk3576.dtsi	rk3576-evb1.dtsi		rk3576-linux.dtsi
rk3576-pinctrl.dtsi	rk3576-evb.dtsi	rk3576-rk806.dtsi	
rockchip-pinconf.dtsi	rk3576-evb-input-keymap.dtsi		

当前默认编译的设备树文件为 tl3576-evm.dts，如需替换为支持 LVDS LCD 显示的设备树，请参考以下步骤修改。

进入 LinuxSDK 源码目录, 执行如下命令, 打开配置文件, 修改内容并保存退出。

Host# vi device/rockchip/rk3576/tl3576_evm_defconfig

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ vi device/rockchip/rk3576/tl3576_evm_defconfig
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$
```

图 22

修改内容如下：

```
#RK_KERNEL_DTS_NAME="tl3576-evm" //注释默认设备树文件 tl3576-evm
RK_KERNEL_DTS_NAME="tl3576-lvds-1024x768" //添加设备树文件 tl3576-lvds-1024x768
```

```
1 RK_UBOOT_SPL=y
2 #RK_KERNEL_DTS_NAME="tl3576-evm"
3 RK_KERNEL_DTS_NAME="tl3576-lvds-1024x768"
4 RK_USE_FIT_IMG=y
```

图 23

执行如下命令，重新配置 LinuxSDK。

Host# ./build.sh lunch:tl3576_evm_defconfig

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ ./build.sh lunch:tl3576_evm_defconfig
Log colors: message notice warning error fatal
Log saved at /home/tronlong/RK3576/rk3576_linux6.1_release/output/sessions/2024-12-30_17-55-56
Switching to defconfig: /home/tronlong/RK3576/rk3576_linux6.1_release/device/rockchip/.chip/tl3576_evm_defconfig
#
# configuration written to /home/tronlong/RK3576/rk3576_linux6.1_release/output/.config
#
Using last kernel version(6.1)
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$
```

图 24

2.1.3 配置内核选项

我司默认配置的是 Linux 内核（推荐使用）。如需修改内核编译选项配置，请继续执行"make xxx menuconfig"命令打开图形配置界面。此命令通过读取当前目录的.config 文件，以图形界面的方式提供内核编译配置查看和修改方法。如无需修改内核编译选项配置，可跳过此步骤。

menuconfig 是一套图形化的配置工具，需要 ncurses 库支持。ncurses 库提供了一系列的 API 函数供调用者生成基于文本的图形界面，因此需要先执行如下命令通过网络安装 ncurses 库。

Host# sudo apt-get install libncurses5-dev

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ sudo apt-get install libncurses5-dev
[sudo] password for tronlong:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
libncurses5-dev is already the newest version (6.3-2ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 44 not upgraded.
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$
```

图 25

Host# sudo apt-get install libncursesw5-dev

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ sudo apt-get install libncursesw5-dev
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
libncursesw5-dev is already the newest version (6.3-2ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 44 not upgraded.
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$
```

图 26

ncurses 库安装完成后，执行如下命令打开图形配置界面。

Host# ./build.sh kernel-config

因我们的存在，让嵌入式应用更简单


```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ ./build.sh kernel-config
Log colors: message notice warning error fatal

Log saved at /home/tronlong/RK3576/rk3576_linux6.1_release/output/sessions/2024-12-30_17-57-23
Using last kernel version(6.1)

=====
Final configs
=====
RK_BOOT_FIT_ITS=/home/tronlong/RK3576/rk3576_linux6.1_release/device/rockchip/.chip/boot.its
RK_BOOT_IMG=boot.img
RK_BUILDROOT=y
RK_BUILDROOT_CFG=rockchip_rk3576
RK_CHIP=rk3576
RK_CHIP_FAMILY=rk3576
RK_DEBIAN=y
RK_DEBIAN_ARCH=arm64
RK_DEBIAN_BOOKWORM=y
RK_DEBIAN_MIRROR=mirrors.ustc.edu.cn
RK_DEBIAN_VERSION=bookworm
RK_DEFCONFIG=tl3576_evm_defconfig
RK_EXTRA_PARTITION_NUM=2
RK_EXTRA_PARTITION_STR=oem:oem:/oem:ext4:defaults:normal:auto:@userdata:userdata:/userdata:ext4:defaults:normal:auto:@@@
RK_KERNEL=y
RK_KERNEL_ARCH=arm64
RK_KERNEL_CFG=rockchip_linux_defconfig
RK_KERNEL_DTS=kernel/arch/arm64/boot/dts/rockchip/tl3576-evm.dts
RK_KERNEL_IMG=kernel/arch/arm64/boot/Image
```

图 27

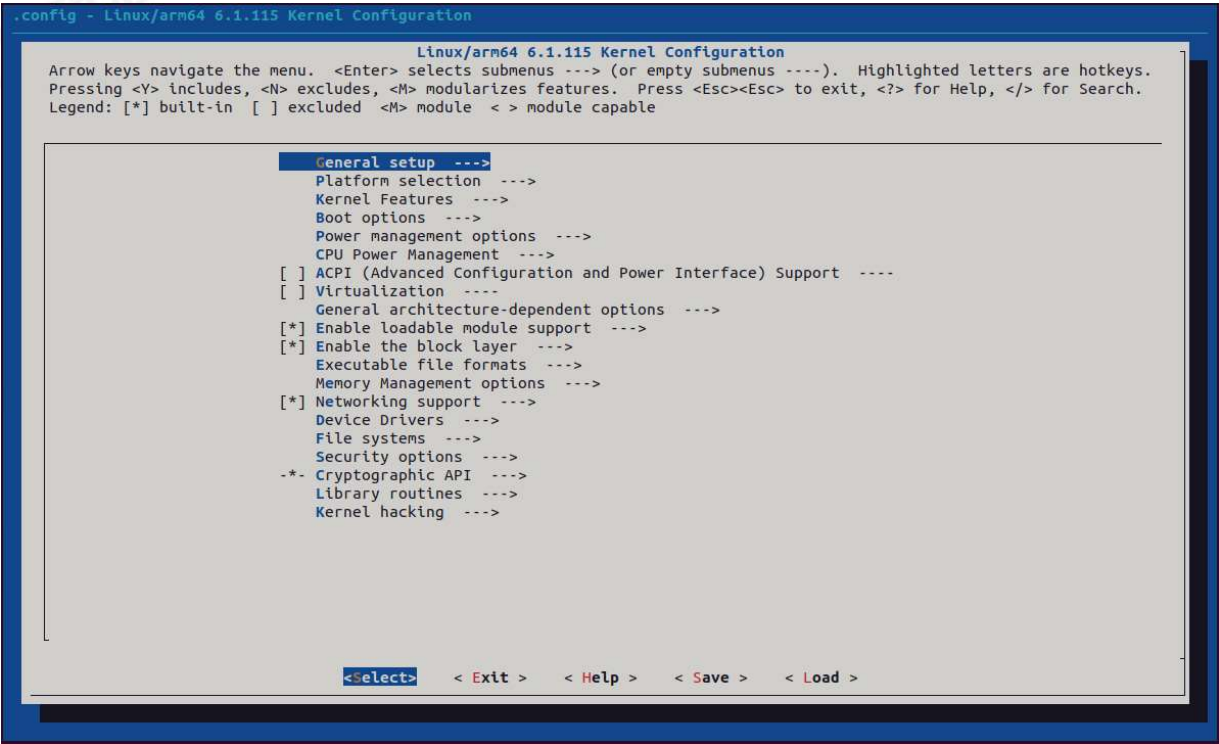


图 28

可通过键盘的方向键选中对应菜单栏。在<Select>被选中的情况下，可按 **Enter** 键进入子菜单。菜单选项中蓝色高亮的字母代表此菜单选项的快捷键，可在键盘上按下对应

因我们的存在，让嵌入式应用更简单

的字母快速选中对应的菜单选项。

每个菜单选项前的括号内容表示当前菜单选项的配置状态。选中对应的菜单选项后，按下 **Y** 键，会将相应的选项配置编译到内核中，同时菜单选项前面变为<*>。按下 **N** 键，不会将相应的选项配置编译到内核中。按下 **M** 键，会将相应的选项配置编译为内核模块，菜单选项前面变为< M>。如需搜索，可按下/键打开搜索框，然后输入要搜索的内容。

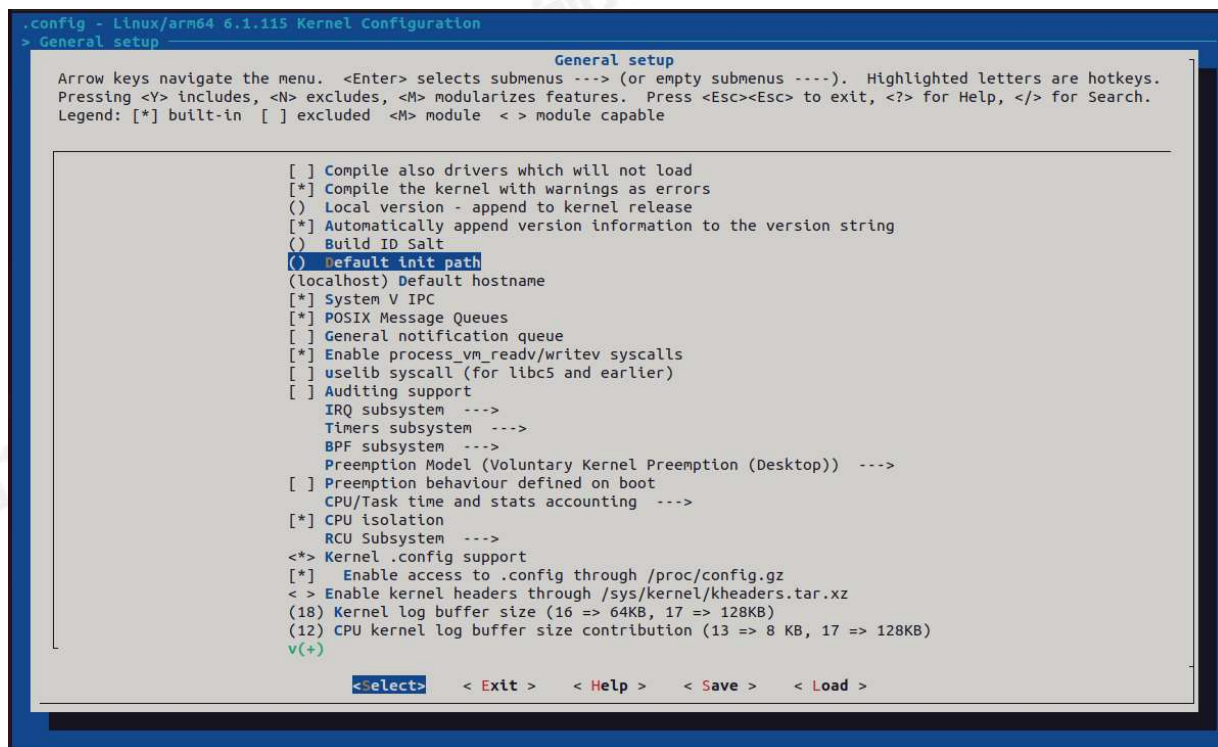


图 29

配置完毕后，选中<Save>，按 Enter 键保存配置选项。然后选中<Exit>，按 Enter 键退出。配置项将自动保存到"kernel/arch/arm64/configs/rockchip_linux_defconfig"。

如需配置为 Linux-RT 内核，请先执行执行如下命令备份 rockchip_linux_defconfig 文件。

```
Host# cd kernel
```

```
Host# cp arch/arm64/configs/rockchip_linux_defconfig arch/arm64/configs/rockchip_linux_defconfig-bak
```

因我们的存在，让嵌入式应用更简单

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ cd kernel
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release/kernel$ cp arch/arm64/configs/rockchip_linux_defconfig arch/arm64/
configs/rockchip_linux_defconfig-bak
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release/kernel$
```

图 30

执行如下命令，在 rockchip_linux_defconfig 基础上添加 rockchip_rt.config 的配置。

Host# make ARCH=arm64 rockchip_linux_defconfig rockchip_rt.config

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release/kernel$ make ARCH=arm64 rockchip_linux_defconfig rockchip_rt.config
ig
#
# configuration written to .config
#
Using .config as base
Merging ./arch/arm64/configs/rockchip_rt.config
Value of CONFIG_ARM_PSCI_CPUIDLE is redefined by fragment ./arch/arm64/configs/rockchip_rt.config:
Previous value: CONFIG_ARM_PSCI_CPUIDLE=y
New value: # CONFIG_ARM_PSCI_CPUIDLE is not set

Value of CONFIG_ARM_ROCKCHIP_DMC_DEVFREQ is redefined by fragment ./arch/arm64/configs/rockchip_rt.config:
Previous value: CONFIG_ARM_ROCKCHIP_DMC_DEVFREQ=y
New value: # CONFIG_ARM_ROCKCHIP_DMC_DEVFREQ is not set

Value of CONFIG_CGROUP_CPUACCT is redefined by fragment ./arch/arm64/configs/rockchip_rt.config:
Previous value: CONFIG_CGROUP_CPUACCT=y
New value: # CONFIG_CGROUP_CPUACCT is not set

Value of CONFIG_CGROUP_SCHED is redefined by fragment ./arch/arm64/configs/rockchip_rt.config:
Previous value: CONFIG_CGROUP_SCHED=y
New value: # CONFIG_CGROUP_SCHED is not set

Value of CONFIG_NO_HZ_FULL is redefined by fragment ./arch/arm64/configs/rockchip_rt.config:
Previous value: # CONFIG_NO_HZ_FULL is not set
New value: CONFIG_NO_HZ_FULL=y
```

图 31

```
Value of CONFIG_CRYPTO_AUTHENC is redefined by fragment ./arch/arm64/configs/rockchip_rt.config:
Previous value: CONFIG_CRYPTO_AUTHENC=y
New value: CONFIG_CRYPTO_AUTHENC=n

Value of CONFIG_CRYPTO_SEQIV is redefined by fragment ./arch/arm64/configs/rockchip_rt.config:
Previous value: CONFIG_CRYPTO_SEQIV=y
New value: CONFIG_CRYPTO_SEQIV=n

Value of CONFIG_CRYPTO_ECHAINIV is redefined by fragment ./arch/arm64/configs/rockchip_rt.config:
Previous value: CONFIG_CRYPTO_ECHAINIV=y
New value: CONFIG_CRYPTO_ECHAINIV=n

#
# merged configuration written to .config (needs make)
#
.config:8072:warning: override: NO_HZ_FULL changes choice state
.config:8089:warning: override: HZ_PERIODIC changes choice state
.config:8090:warning: override: HZ_1000 changes choice state
.config:8091:warning: override: PREEMPT_RT changes choice state
#
# configuration written to .config
#
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release/kernel$
```

图 32

请执行如下命令，保存配置选项至 defconfig 文件，再将 defconfig 文件内容覆盖至 rockchip_linux_defconfig 文件。

Host# make ARCH=arm64 savedefconfig

因我们的存在，让嵌入式应用更简单

Host# mv defconfig arch/arm64/configs/rockchip_linux_defconfig

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release/kernel$ make ARCH=arm64 savedefconfig
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release/kernel$ mv defconfig arch/arm64/configs/rockchip_linux_defconfig
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release/kernel$
```

图 33

若编译 Linux-RT 内核后，如需恢复使用默认内核配置，可执行如下命令进行恢复，然后参考“局部编译”章节进行内核编译即可。

Host# cd kernel

Host# mv arch/arm64/configs/rockchip_linux_defconfig-bak arch/arm64/configs/rockchip_linux_defconfig

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ cd kernel
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release/kernel$ mv arch/arm64/configs/rockchip_linux_defconfig-bak arch/a
rm64/configs/rockchip_linux_defconfig
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release/kernel$
```

图 34

2.1.4 配置 Buildroot 编译选项

如需修改 Buildroot 配置，请执行如下命令打开 Buildroot 配置界面。若无需修改 Buildroot 配置，可跳过此步骤。

Host# cd /home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/

Host# ./envsetup.sh rockchip_rk3576

Host# make menuconfig

```
tronlong@tronlong-virtual-machine:~$ cd /home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release/buildroot$ ./envsetup.sh rockchip_rk3576

##### Rockchip Buildroot SDK #####

make: Entering directory '/home/tronlong/RK3576/rk3576_linux6.1_release/buildroot'
mkdir -p /home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/build/buildroot-config/lxdialog
PKG_CONFIG_PATH="" make CC="/usr/bin/gcc" HOSTCC="/usr/bin/gcc" \
  obj=/home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/build/buildroot-config -C support/k
config -f Makefile.br conf
/usr/bin/gcc -D_DEFAULT_SOURCE -D_XOPEN_SOURCE=600 -DCURSES_LOC="ncurses.h" -DNCURSES_WIDECHAR=1 -DLOCALE -I/home/tronl
ong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/build/buildroot-config -DCONFIG="" -MM *.c > /home
/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/build/buildroot-config/.depend 2>/dev/null || :
/usr/bin/gcc -D_DEFAULT_SOURCE -D_XOPEN_SOURCE=600 -DCURSES_LOC="ncurses.h" -DNCURSES_WIDECHAR=1 -DLOCALE -I/home/tronl
ong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/build/buildroot-config -DCONFIG="" -c conf.c -o /
home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/build/buildroot-config/conf.o
/usr/bin/gcc -D_DEFAULT_SOURCE -D_XOPEN_SOURCE=600 -DCURSES_LOC="ncurses.h" -DNCURSES_WIDECHAR=1 -DLOCALE -I/home/tronl
ong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/build/buildroot-config -DCONFIG="" -I. -c /home/tr
onlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/build/buildroot-config/zconf.tab.c -o /home/tronlon
g/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/build/buildroot-config/zconf.tab.o
/usr/bin/gcc -D_DEFAULT_SOURCE -D_XOPEN_SOURCE=600 -DCURSES_LOC="ncurses.h" -DNCURSES_WIDECHAR=1 -DLOCALE -I/home/tronl
ong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/build/buildroot-config -DCONFIG="" /home/tronlong
/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/build/buildroot-config/zconf.tab.o -o /home/tronlong/RK3576/rk3576_li
nux6.1_release/buildroot/output/rockchip_rk3576/build/buildroot-config/zconf.tab.o
rm /home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/build/buildroot-config/zconf.tab.c
GEN /home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/Makefile
Parsing defconfig: /home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/configs/rockchip_rk3576_defconfig
Using /home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/configs/rockchip_rk3576_defconfig as base
#
# merged configuration written to /home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/.config.i
n (needs make)
#
#
# configuration written to /home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/.config
#
make: Leaving directory '/home/tronlong/RK3576/rk3576_linux6.1_release/buildroot'
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release/buildroot$
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release/buildroot$ make menuconfig
```

图 35

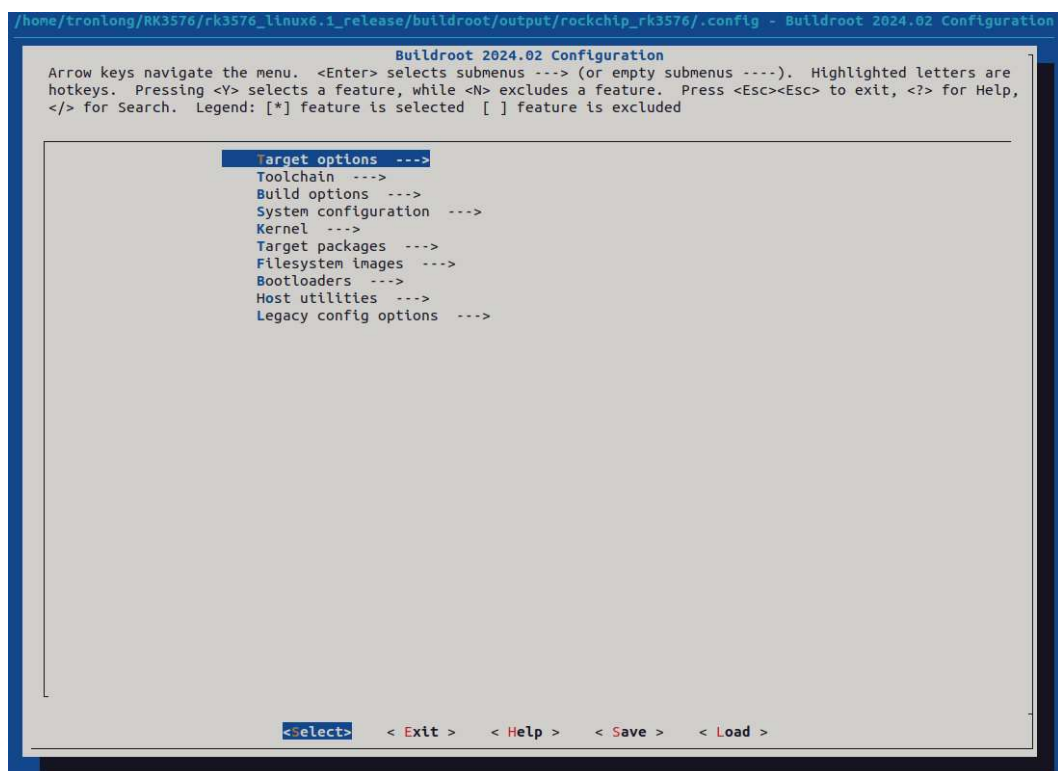


图 36

可通过键盘的方向键选中对应菜单栏。在<Select>被选中的情况下，可按 **Enter** 键进
因我们的存在，让嵌入式应用更简单

入子菜单。菜单选项中蓝色高亮的字母代表此菜单选项的快捷键，可在键盘上按下对应的字母快速选中对应的菜单选项。

每个菜单选项前的括号内容表示当前菜单选项的配置状态。选中对应的菜单选项后，按下 **Y** 键，会将相应的选项配置编译到 **Buildroot** 中，同时菜单选项前面变为 **< * >**。按下 **N** 键，不会将相应的选项配置编译到 **Buildroot** 中。如需搜索，可按下 **/** 键打开搜索框，然后输入要搜索的内容。

配置完毕后，选中 **<Save>**，按 **Enter** 键保存配置选项。然后选中 **<Exit>**，按 **Enter** 键退出。

执行如下命令，保存 **Buildroot** 配置项至 `"/home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/configs/rockchip_rk3576_defconfig"`。

Host# make savedefconfig

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release/buildroot$ make savedefconfig
GEN /home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/Makefile
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release/buildroot$
```

图 37

2.2 编译 LinuxSDK

在 **LinuxSDK** 的 `rk3576_linux6.1_release` 目录下执行如下命令，一键编译生成 **U-Boot**、**Kernel**、**Buildroot**、**Recovery** 等镜像，并打包生成 **Linux** 系统镜像。

Host# ./build.sh


```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ ./build.sh
Log colors: message notice warning error fatal

Log saved at /home/tronlong/RK3576/rk3576_linux6.1_release/output/sessions/2024-12-30_18-27-05
Using last kernel version(6.1)

=====
Final configs
=====
RK_BOOT_FIT_ITS=/home/tronlong/RK3576/rk3576_linux6.1_release/device/rockchip/.chip/boot.its
RK_BOOT_IMG=boot.img
RK_BUILDROOT=y
RK_BUILDROOT_CFG=rockchip_rk3576
RK_CHIP=rk3576
RK_CHIP_FAMILY=rk3576
RK_DEBIAN=y
RK_DEBIAN_ARCH=arm64
RK_DEBIAN_BOOKWORM=y
RK_DEBIAN_MIRROR=mirrors.ustc.edu.cn
RK_DEBIAN_VERSION=bookworm
RK_DEFCONFIG=tl3576_evm_defconfig
RK_EXTRA_PARTITION_NUM=2
RK_EXTRA_PARTITION_STR=oem:oem:/oem:ext4:defaults:normal:auto:@userdata:userdata:/userdata:ext4:defaults:normal:auto:@@@
RK_KERNEL=y
RK_KERNEL_ARCH=arm64
RK_KERNEL_CFG=rockchip_linux_defconfig
RK_KERNEL_DTS=kernel/arch/arm64/boot/dts/rockchip/tl3576-evm.dts
```

图 38

预计耗时约 2 小时后编译完成，最终在 rockdev 目录下生成 U-Boot、Kernel、Buildroot、Recovery、Linux 系统等镜像如下所示。

```
Add file: ./package-file
package-file,Add file: ./package-file done,offset=0x800,size=0xe1,userspace=0x1
Add file: ./parameter.txt
parameter,Add file: ./parameter.txt done,offset=0x1000,size=0x227,userspace=0x1,flash_address=0x00000000
Add file: ./MiniLoaderAll.bin
bootloader,Add file: ./MiniLoaderAll.bin done,offset=0x1800,size=0xbb1f9,userspace=0x177
Add file: ./uboot.img
uboot,Add file: ./uboot.img done,offset=0xbd000,size=0x400000,userspace=0x800,flash_address=0x00004000
Add file: ./misc.img
misc,Add file: ./misc.img done,offset=0x4bd000,size=0xc000,userspace=0x18,flash_address=0x00006000
Add file: ./boot.img
boot,Add file: ./boot.img done,offset=0x4c9000,size=0x2963200,userspace=0x52c7,flash_address=0x00008000
Add file: ./recovery.img
recovery,Add file: ./recovery.img done,offset=0x2e2c800,size=0x3957800,userspace=0x72af,flash_address=0x00028000
Add file: ./rootfs.img
rootfs,Add file: ./rootfs.img done,offset=0x6784000,size=0x5210000,userspace=0xa4200,flash_address=0x00078000
Add file: ./oem.img
oem,Add file: ./oem.img done,offset=0x58884000,size=0x11ef000,userspace=0x23de,flash_address=0x01c78000
Add file: ./userdata.img
userdata,Add file: ./userdata.img done,offset=0x59a73000,size=0x800000,userspace=0x1000,flash_address=0x01cb8000
Add CRC...
Make firmware OK!
----- OK -----
*****rkImageMaker ver 2.29*****
Generating new image, please wait...
Writing head info...
Writing boot file...
Writing firmware...
Generating MDS data...
MDS data generated successfully!
New image generated successfully!

Run 'make edit-package-file' if you want to change the package-file.

Running mk-updateimg.sh - build_updateimg succeeded.
Images under /home/tronlong/RK3576/rk3576_linux6.1_release/output/firmware/ are ready!
Running mk-firmware.sh - build_firmware succeeded.
Running 99-all.sh - build_all succeeded.
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$
```

图 39

因我们的存在，让嵌入式应用更简单

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ ls rockdev/
boot.img  MiniLoaderAll.bin  misc.img  oem.img  parameter.txt  recovery.img  rootfs.img  uboot.img  update.img  userdata.img
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$
```

图 40

表 6 镜像文件说明

boot.img	内核镜像，包含内核和设备树
MiniLoaderAll.bin	DDR 初始化镜像和 U-Boot SPL 镜像
misc.img	misc 镜像
oem.img	oem 镜像，存放用户数据，挂载至文件系统的"/oem"目录
parameter.txt	分区表，update.img 按照此分区表存放各个镜像
recovery.img	recovery 镜像，用于系统升级
rootfs.img	文件系统镜像
uboot.img	U-Boot 镜像
userdata.img	userdata 镜像，挂载至文件系统的"/userdata"目录
update.img	Linux 系统镜像，包含了 MiniLoaderAll.bin、parameter.txt、uboot.img、misc.img、boot.img、recovery.img、rootfs.img、oem.img、userdata.img

请参考《Linux 系统启动卡制作及系统固化》文档，将 Linux 系统镜像文件固化至 Micro SD 卡或 eMMC。

2.3 局部编译

如需单独编译 U-Boot、Linux 内核或文件系统，可参照本章节内容进行局部编译。

2.3.1 U-Boot 编译

请先参考“配置 LinuxSDK 编译选项”章节配置 LinuxSDK。在 LinuxSDK 源码目录下执行如下命令，编译 U-Boot。

```
Host# cd /home/tronlong/RK3576/rk3576_linux6.1_release/
Host# ./build.sh uboot
```

```
tronlong@tronlong-virtual-machine:~$ cd /home/tronlong/RK3576/rk3576_linux6.1_release/
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ ./build.sh kernel
Log colors: message notice warning error fatal

Log saved at /home/tronlong/RK3576/rk3576_linux6.1_release/output/sessions/2025-03-14_12-38-54
Using last kernel version(6.1)

=====
Final configs
=====
RK_BOOT_FIT_ITS=/home/tronlong/RK3576/rk3576_linux6.1_release/device/rockchip/.chip/boot.its
RK_BOOT_IMG=boot.img
RK_BUILDROOT=y
RK_BUILDROOT_CFG=rockchip_rk3576
RK_CHIP=rk3576
RK_CHIP_FAMILY=rk3576
RK_CHIP_HAS_GPU=y
RK_DEBIAN=y
RK_DEBIAN_ARCH=arm64
RK_DEBIAN_BOOKWORM=y
RK_DEBIAN_MIRROR=mirrors.ustc.edu.cn
RK_DEBIAN_VERSION=bookworm
RK_DEFCONFIG=tl3576_evm_defconfig
RK_EXTRA_PARTITION_NUM=2
RK_EXTRA_PARTITION_STR=auto:oem:auto:ext4:defaults:normal:auto:@auto:userdata:auto:ext4:defaults:normal:auto:@@
RK_KERNEL=y
RK_KERNEL_ARCH=arm64
RK_KERNEL_CFG=rockchip_linux_defconfig
RK_KERNEL_DTS=kernel/arch/arm64/boot/dts/rockchip/tl3576-evm.dts
RK_KERNEL_IMG=kernel/arch/arm64/boot/Image
```

图 41

```
*****boot_merger ver 1.35*****
Info:Pack loader ok.
creating new idblock from loader...
idblock binary saving at rk3576_idblock_v1.09.107.img
pack loader(SPL) okay! Input: /home/tronlong/RK3576/rk3576_linux6.1_release/rkbin/RKBOOT/RK3576MINIAL.INI

/home/tronlong/RK3576/rk3576_linux6.1_release/u-boot
pack loader with new: spl/u-boot-spl.bin

Image(no-signed, version=0): uboot.img (FIT with uboot, trust...) is ready
Image(no-signed): rk3576_idblock_v1.09.107.img (with spl, ddr...) is ready
pack uboot.img okay! Input: /home/tronlong/RK3576/rk3576_linux6.1_release/rkbin/RKTRUST/RK3576TRUST.INI

Platform RK3576 is build OK, with new .config(make rk3576_defconfig -j16)
/home/tronlong/RK3576/rk3576_linux6.1_release/prebuilts/gcc/linux-x86/aarch64/gcc-arm-10.3-2021.07-x86_64-aarch64-none-linux-gnu/bin/
/aarch64-none-linux-gnu-
Fri Mar 14 10:16:22 CST 2025
+ cd ..
Running 70-loader.sh - build_uboot succeeded.
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$
```

图 42

编译完成后，在 rockdev 目录下生成 MiniLoaderAll.bin 文件和镜像 uboot.img。

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ ls rockdev/MiniLoaderAll.bin
rockdev/MiniLoaderAll.bin
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ ls rockdev/uboot.img
rockdev/uboot.img
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$
```

图 43

2.3.2 内核编译

请先参考“配置 LinuxSDK 编译选项”章节配置 LinuxSDK。在 LinuxSDK 源码目录下执行如下命令，编译内核。

因我们的存在，让嵌入式应用更简单

Host# cd /home/tronlong/RK3576/rk3576_linux6.1_release/

Host# ./build.sh kernel

```
tronlong@tronlong-virtual-machine:~$ cd /home/tronlong/RK3576/rk3576_linux6.1_release/
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ ./build.sh kernel
Log colors: message notice warning error fatal

Log saved at /home/tronlong/RK3576/rk3576_linux6.1_release/output/sessions/2025-03-14_12-39-48
Using last kernel version(6.1)

=====
Final configs
=====
RK_BOOT_FIT_ITS=/home/tronlong/RK3576/rk3576_linux6.1_release/device/rockchip/.chip/boot.its
RK_BOOT_IMG=boot.img
RK_BUILDROOT=y
RK_BUILDROOT_CFG=rockchip_rk3576
RK_CHIP=rk3576
RK_CHIP_FAMILY=rk3576
RK_CHIP_HAS_GPU=y
RK_DEBIAN=y
RK_DEBIAN_ARCH=arm64
RK_DEBIAN_BOOKWORM=y
RK_DEBIAN_MIRROR=mirrors.ustc.edu.cn
RK_DEBIAN_VERSION=bookworm
RK_DEFCONFIG=tl3576_evm_defconfig
RK_EXTRA_PARTITION_NUM=2
RK_EXTRA_PARTITION_STR=auto:oem:auto:ext4:defaults:normal:auto:@auto:userdata:auto:ext4:defaults:normal:auto:@@@
RK_KERNEL=y
RK_KERNEL_ARCH=arm64
RK_KERNEL_CFG=rockchip_linux_defconfig
RK_KERNEL_DTS=kernel/arch/arm64/boot/dts/rockchip/tl3576-evm.dts
RK_KERNEL_IMG=kernel/arch/arm64/boot/Image
```

图 44

```
Created:      Fri Mar 14 10:59:45 2025
Image 0 (fdt)
Description:  unavailable
Created:      Fri Mar 14 10:59:45 2025
Type:         Flat Device Tree
Compression:  uncompressed
Data Size:    286679 Bytes = 279.96 KiB = 0.27 MiB
Architecture: AArch64
Load Address: 0xffffffff00
Hash algo:    sha256
Hash value:   d481782a47dd1c556abaacbfd188d04787658368ab4ba4e5dc1a6554c08bb842
Image 1 (kernel)
Description:  unavailable
Created:      Fri Mar 14 10:59:45 2025
Type:         Kernel Image
Compression:  uncompressed
Data Size:    42795520 Bytes = 41792.50 KiB = 40.81 MiB
Architecture: AArch64
OS:           Linux
Load Address: 0xffffffff01
Entry Point:  0xffffffff01
Hash algo:    sha256
Hash value:   3a7148ce7250ee2861c68233c86f58d6e5c3be65216a4743d20b479a79239b5b
Image 2 (resource)
Description:  unavailable
Created:      Fri Mar 14 10:59:45 2025
Type:         Multi-File Image
Compression:  uncompressed
Data Size:    324608 Bytes = 317.00 KiB = 0.31 MiB
Hash algo:    sha256
Hash value:   83b3b3662c9312987b5e3cc3a3b5b5aef0c0266061306d7db122285cc5ae7480
Default Configuration: 'conf'
Configuration 0 (conf)
Description:  unavailable
Kernel:      kernel
FDT:         fdt
+ ln -rsf kernel/boot.img /home/tronlong/RK3576/rk3576_linux6.1_release/output/firmware/boot.img
Not Found io-domains in kernel/arch/arm64/boot/dts/rockchip/tl3576-evm.dts
Running 10-kernel.sh - build_kernel succeeded.
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$
```

图 45

编译完成后，在 rockdev 目录下生成镜像 boot.img。

因我们的存在，让嵌入式应用更简单

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ ls rockdev/boot.img
rockdev/boot.img
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$
```

图 46

2.3.3 文件系统编译

在 LinuxSDK 源码目录下执行如下命令，编译文件系统。

备注：编译文件系统前，请确保已参考“解压 dl.tar.gz 压缩包”章节解压 dl.tar.gz 压缩包至 LinuxSDK 的"rk3576_linux6.1_release/buildroot/"目录。

Host# ./build.sh buildroot

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ ./build.sh buildroot
Log colors: message notice warning error fatal

Log saved at /home/tronlong/RK3576/rk3576_linux6.1_release/output/sessions/2025-03-14_11-00-59
Using last kernel version(6.1)
Using rootfs system(buildroot) from cmdline

=====
Final configs
=====
RK_BOOT_FIT_ITS=/home/tronlong/RK3576/rk3576_linux6.1_release/device/rockchip/.chip/boot.its
RK_BOOT_IMG=boot.img
RK_BUILDROOT=y
RK_BUILDROOT_CFG=rockchip_rk3576
RK_CHIP=rk3576
RK_CHIP_FAMILY=rk3576
RK_CHIP_HAS_GPU=y
RK_DEBIAN=y
RK_DEBIAN_ARCH=arm64
RK_DEBIAN_BOOKWORM=y
RK_DEBIAN_MIRROR=mirrors.ustc.edu.cn
RK_DEBIAN_VERSION=bookworm
RK_DEFCONFIG=tl3576_evm_defconfig
RK_EXTRA_PARTITION_NUM=2
RK_EXTRA_PARTITION_STR=auto:oem:auto:ext4:defaults:normal:auto:@auto:userdata:auto:ext4:defaults:normal:auto:@@
RK_KERNEL=y
RK_KERNEL_ARCH=arm64
RK_KERNEL_CFG=rockchip_linux_defconfig
RK_KERNEL_DTS=kernel/arch/arm64/boot/dts/rockchip/tl3576-evm.dts
RK_KERNEL_IMG=kernel/arch/arm64/boot/Image
```

图 47

```
Running 95-extra-parts.sh for /home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/target (buildroot init=
busybox)...
Preparing extra partitions...
Running 97-ldcache.sh for /home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/target (buildroot init=busy
box)...
Creating ld.so.cache for ...
/home/tronlong/RK3576/rk3576_linux6.1_release/device/rockchip/common/tools/x86_64/ldconfig: Can't open configuration file /home/tron
long/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/target/etc/ld.so.conf: No such file or directory
/home/tronlong/RK3576/rk3576_linux6.1_release/device/rockchip/common/tools/x86_64/ldconfig: /lib/libstdc++.so.6.0.30-gdb.py is not a
n ELF file - it has the wrong magic bytes at the start.

saving cache in little endian encoding
Running 99-owner.sh for /home/tronlong/RK3576/rk3576_linux6.1_release/buildroot/output/rockchip_rk3576/target (buildroot init=busybo
x)...
Running 30-rootfs.sh - build_buildroot /home/tronlong/RK3576/rk3576_linux6.1_release/output/buildroot/images succeeded.
Running 30-rootfs.sh - build_rootfs buildroot succeeded.
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$
```

图 48

因我们的存在，让嵌入式应用更简单

编译完成后，在 rockdev 目录下生成镜像 rootfs.img。

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ ls rockdev/rootfs.img
rockdev/rootfs.img
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$
```

图 49

2.3.4 生成 update.img 镜像

单独编译 U-Boot、内核、文件系统后，可以将所有镜像重新打包，重新生成 update.img。

备注：如需替换 U-Boot、内核、文件系统镜像，可替换"rk3576_linux6.1_release/output/firmware/"目录下对应的镜像，重新生成 update.img。

进入 LinuxSDK 源码目录，执行如下命令，重新生成 update.img。

Host# ./build.sh firmware

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ ./build.sh firmware
Log colors: message notice warning error fatal

Log saved at /home/tronlong/RK3576/rk3576_linux6.1_release/output/sessions/2025-03-14_12-40-26
Using last kernel version(6.1)

=====
Final configs
=====
RK_BOOT_FIT_ITS=/home/tronlong/RK3576/rk3576_linux6.1_release/device/rockchip/.chip/boot.its
RK_BOOT_IMG=boot.img
RK_BUILDROOT=y
RK_BUILDROOT_CFG=rockchip_rk3576
RK_CHIP=rk3576
RK_CHIP_FAMILY=rk3576
RK_CHIP_HAS_GPU=y
RK_DEBIAN=y
RK_DEBIAN_ARCH=arm64
RK_DEBIAN_BOOKWORM=y
RK_DEBIAN_MIRROR=mirrors.ustc.edu.cn
RK_DEBIAN_VERSION=bookworm
RK_DEFCONFIG=tl3576_evm_defconfig
RK_EXTRA_PARTITION_NUM=2
RK_EXTRA_PARTITION_STR=auto:oem:auto:ext4:defaults:normal:auto:@auto:userdata:auto:ext4:defaults:normal:auto:@@@
RK_KERNEL=y
RK_KERNEL_ARCH=arm64
RK_KERNEL_CFG=rockchip_linux_defconfig
RK_KERNEL_DTS=kernel/arch/arm64/boot/dts/rockchip/tl3576-evm.dts
RK_KERNEL_IMG=kernel/arch/arm64/boot/Image
```

图 50

编译完成后，在 rockdev 目录下生成镜像 update.img。

```
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$ ls rockdev/update.img
rockdev/update.img
tronlong@tronlong-virtual-machine:~/RK3576/rk3576_linux6.1_release$
```

图 51

因我们的存在，让嵌入式应用更简单